Practice workbook · Read or estimate a value from a graph, chart, table or image

Locating a required value, year or feature on a graph or chart, adding readings to a table, or estimating a proportion or size by comparing lengths in an image.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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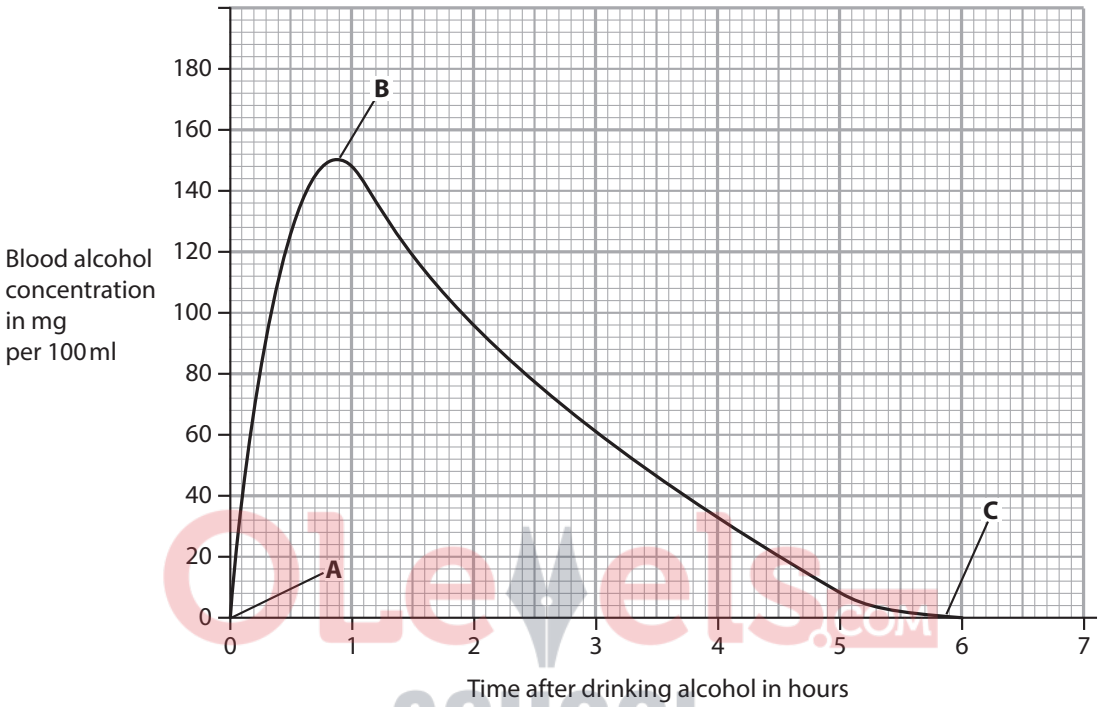
Type ID: HBI-MATH-READ-A-VALUE

1. Question 6 · part b.i · 1 mark

June 2025 Paper 1

Worked-example retry · Attempt part b.i only.

6 The graph shows the changes in blood alcohol concentration of a person who drinks alcohol.



(a) (i) Explain the shape of the graph between points A and B.

(2)

(ii) Explain the shape of the graph between points B and C.

(3)

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1. Question 6 · part b.i · 1 mark

June 2025 Paper 1

Worked-example retry · Attempt part b.i only.

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(b) In the United Kingdom the legal limit to drive is less than 80 mg per 100 ml of alcohol in the blood.

(i) The person shown in the graph drinks alcohol and the alcohol level in the blood goes above the legal limit to drive.

Determine how many hours after drinking alcohol it would be legal for the person shown in the graph to drive.

(1)

(ii) Explain why it is against the law to drive with high levels of alcohol in the blood.

(4)

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1. Question 6 · part b.i · 1 mark · Working and answer

Working and answer

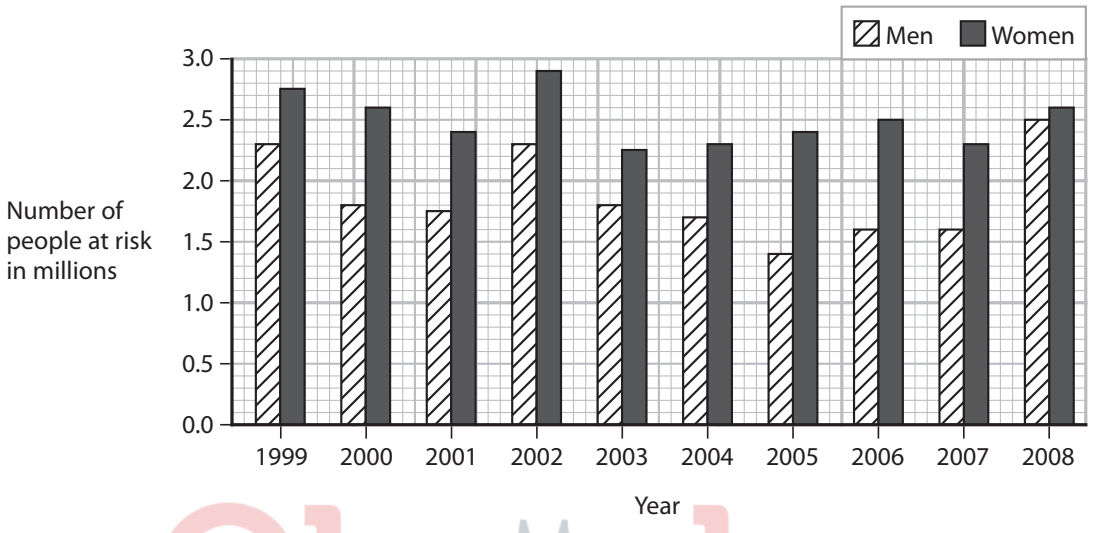
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2. Question 2 · part a.i · 1 mark

June 2022 Paper 2

Further practice · Attempt part a.i only.

2 The graph shows the number of working-age people at risk of developing a mental illness.



(a) (i) Determine the year when the greatest number of women were at risk of developing a mental illness.

(1)

.....

(ii) Determine the year when there was the greatest difference between the number of males and the number of females at risk of developing a mental illness.

(1)

.....

(iii) Determine the ratio of females at risk of developing a mental illness in 1999 compared with 2008.

Give your answer in the form n : 1

(2)

ratio = ...

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2. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

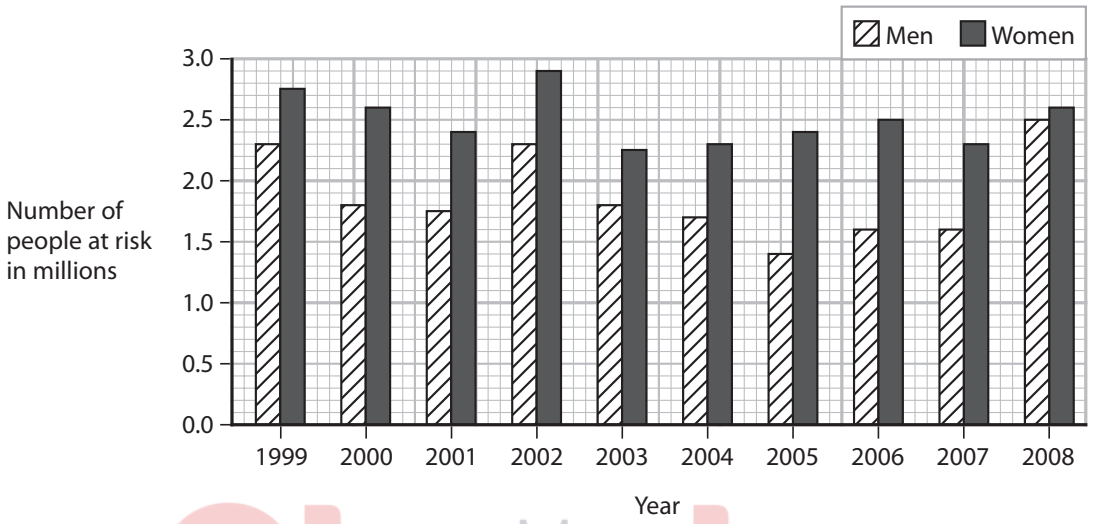
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3. Question 2 · part a.ii · 1 mark

June 2022 Paper 2

Further practice · Attempt part a.ii only.

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(2)

ratio = ...

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3. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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4. Question 8 · part c.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part c.i only.

8 (a) When women cannot conceive naturally, they can have fertility treatment.

Women having fertility treatment can be given the hormones FSH and LH.

(i) Give a role of LH in women.

(1)

(ii) Giving birth to twins or triplets is described as a multiple birth.

Explain how the use of FSH to treat infertility could increase the chances of a multiple birth.

(3)

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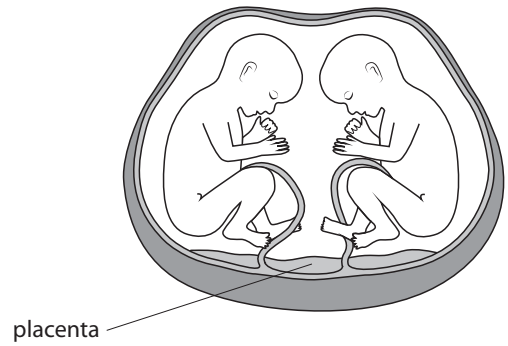
4. Question 8 · part c.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part c.i only.

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(b) Some identical twins share a placenta, as shown in the diagram.



(i) Twins that share a placenta are often born with a lower mass than twins that have separate placentas.

Explain why twins sharing a placenta have a lower mass when they are born.

(3)

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(ii) Identical twins form when the embryo splits early in pregnancy.

Explain why the twins are genetically identical.

(2)

4. Question 8 · part c.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part c.i only.

(c) When babies are young, their growth is monitored using a percentile chart.

The diagram shows a percentile chart for baby girls.



(i) Determine the growth percentile of a six-week-old baby girl with a mass of 5 kg.

(1)

percentile = ...

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4. Question 8 · part c.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part c.i only.

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(ii) Describe how percentile charts are used to identify concerns about the growth of a baby.

(3)

(Total for Question 8 = 13 marks)

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Working and answer

5. Question 4 · part c.i · 1 mark

June 2023 Paper 1

Further practice · Attempt part c.i only.

4 The eye contains receptors that can detect light.

(a) Name two other senses.

(2)

1 .....

2 .....

(b) The table shows how the focal length of the eye lens changes as the thickness of the lens changes.

Thickness of lens in mm	Focal length in mm
10	12.5
9	14.0
8	15.5
7	18.0
6	21.0
5	25.0

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(i) Describe the relationship between the thickness of the lens and its focal length.

(2)

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(ii) Give the effect on the diameter of the lens as its thickness increases.

(1)

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5. Question 4 · part c.i · 1 mark

June 2023 Paper 1

Further practice · Attempt part c.i only.

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(iii) A person is looking at a distant object and then looks at a near object.

Use the data in the table to explain the changes that occur in the eye so the person is able to form an image of the near object on the retina.

(4)

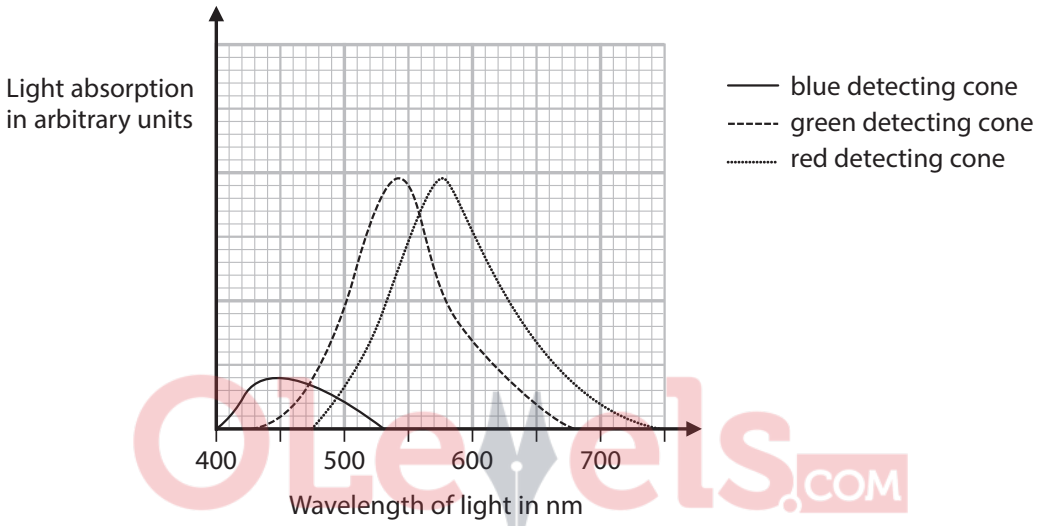
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5. Question 4 · part c.i · 1 mark

June 2023 Paper 1

Further practice · Attempt part c.i only.

(c) Cone cells are found in the retina of the eye. They detect colour. The graph shows the absorption of different wavelengths of light by three types of cone cell.



(i) Determine the wavelength of light at which a red detecting cone absorbs most light.

(1)

wavelength of light = ... nm

(ii) Determine the maximum difference in the wavelengths of the light that can be absorbed by blue detecting cones.

(2)

difference in wavelengths = ... nm

(iii) Suggest why it is better to have three types of cone rather than just one type of cone.

(2)

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(Total for Question 4 = 14 marks)

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5. Question 4 · part c.i · 1 mark · Working and answer

Working and answer

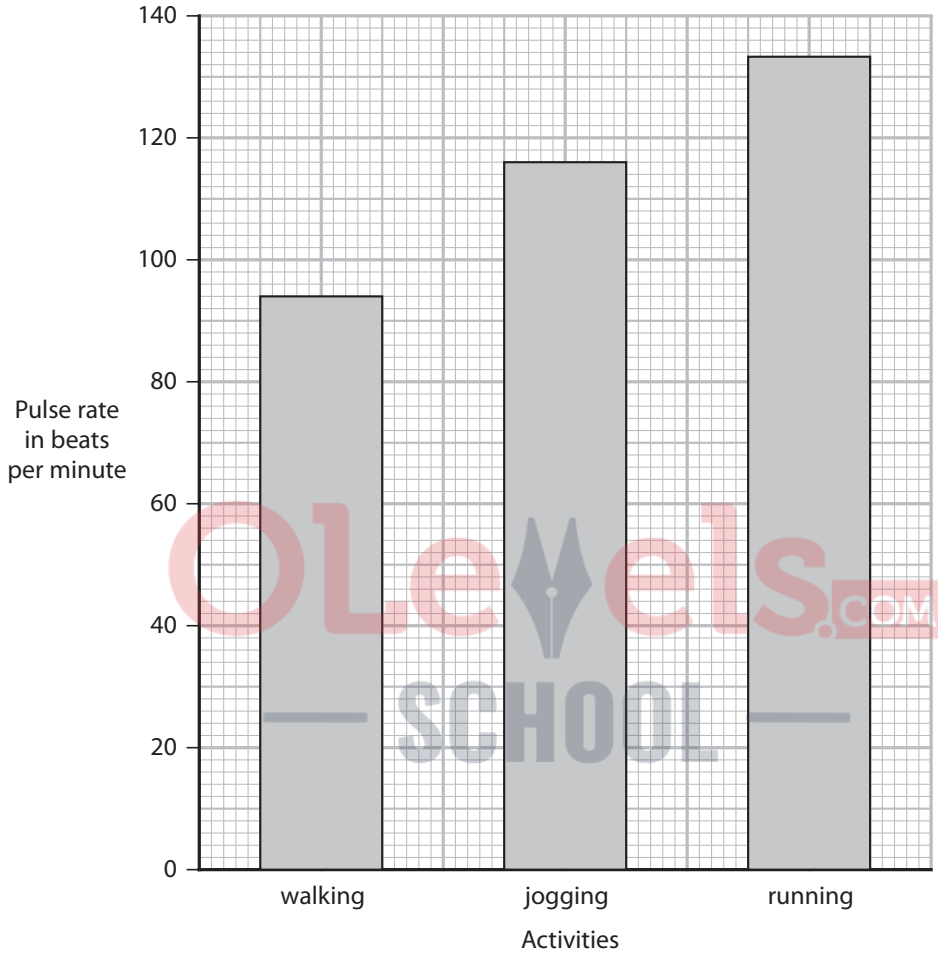
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6. Question 4 · part a.ii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part a.ii only.

4 (a) The bar chart shows the pulse rate of a student at the end of three different activities.



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6. Question 4 · part a.ii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part a.ii only.

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(i) Draw a table that the student could use to record these results.

(3)

(ii) Complete the table by adding the student's results.

(2)

(iii) Describe a method that the student could use to obtain the results shown in the bar chart.

(4)

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6. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

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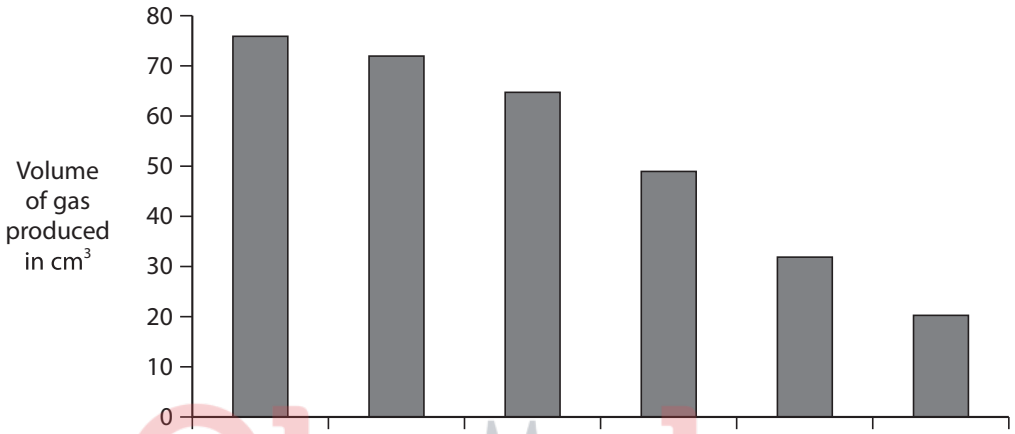
7. Question 3 · part a.ii · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.ii only.

3 A student investigates a reaction. The reaction is catalysed by an enzyme and produces a gas. The student collects the gas for 10 minutes and measures the volume of the gas. The student repeats this every 10 minutes for one hour.

The bar chart shows the student's results.



(a) (i) Add a suitable labelled scale to the *x*-axis.

(2)

(ii) Estimate the total volume of gas produced after 30 minutes.

(2)

volume of gas = ... cm^3

(iii) Describe the trend shown by the results.

(3)

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7. Question 3 · part a.ii · 2 marks · Working and answer

Working and answer

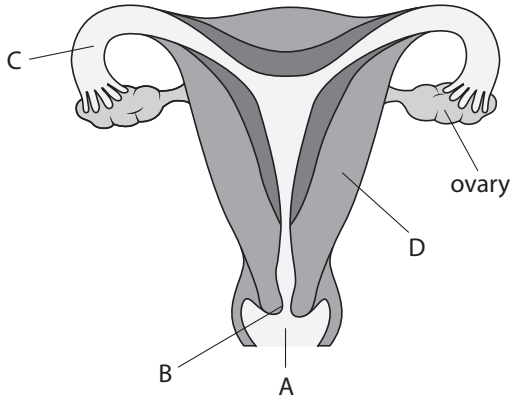
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8. Question 2 · part d.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part d.i only.

2 (a) The diagram shows some structures of the female reproductive system.



(i) Where are sperm deposited during sexual intercourse?

(1)

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

(ii) Which part of the female reproductive system contracts during birth to push the fetus downwards?

(1)

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

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8. Question 2 · part d.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part d.i only.

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(iii) The ovaries produce female sex hormones.

The box lists words associated with the female reproductive system.

FSH	wall	fertilisation	oestrogen	ovulation
lining	ADH	follicles	menstruation	

Use words from the box to complete the passage about the ovaries.

(3)

The ovaries produce a hormone called ..... .

This hormone repairs the uterus ..... following menstruation. Once a month, one egg is released from the ovaries. This is known as ..... .

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(b) State why a pregnant female should have more calcium in her diet than a female who is not pregnant.

(1)

.....

.....

(c) Some women breastfeed their babies.

Give one advantage of breastfeeding babies rather than feeding them bottled milk.

(1)

.....

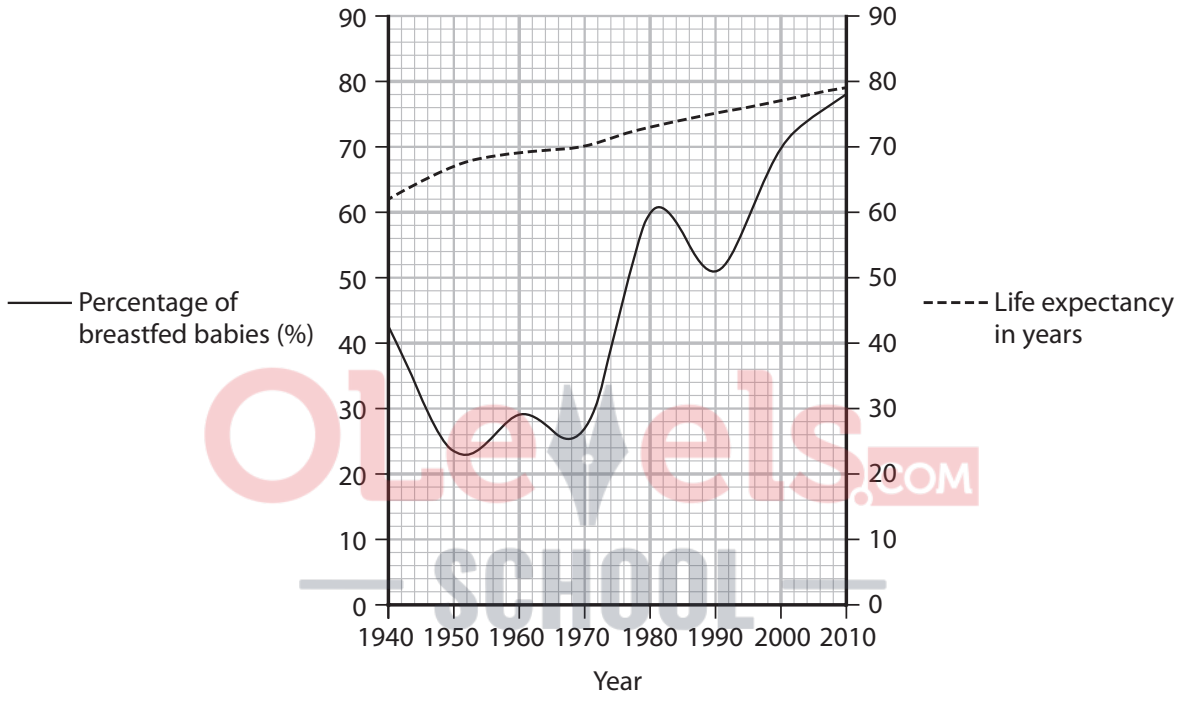
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8. Question 2 · part d.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part d.i only.

(d) The graph shows the percentage of babies that are breastfed and the life expectancy of breastfed babies from 1940 to 2010 in the UK.



(i) Compare the percentage of babies that were breastfed in 1940 with the percentage of babies that were breastfed in 2010.

(1)

(ii) The graph suggests that breastfeeding has no effect on life expectancy.

Describe how the graph shows this.

(2)

(Total for Question 2 = 10 marks)

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8. Question 2 · part d.i · 1 mark · Working and answer

Working and answer

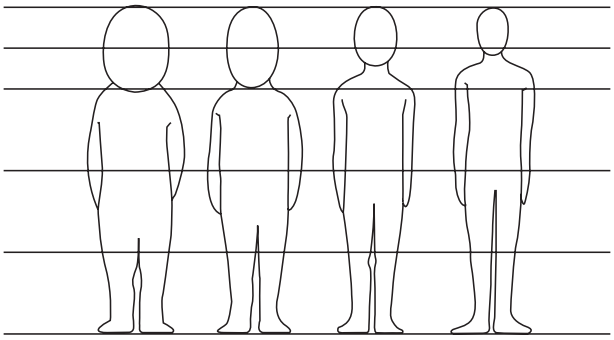
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9. Question 4 · part a.i · 1 mark

June 2019 Paper 2R

Further practice · Attempt part a.i only.

4 The diagram shows the changes in the proportions of different parts of the body of a person over a period of 18 years.



newborn 2 years 7 years 18 years

(a) (i) Estimate the proportion of the body that is the head in a newborn baby.

(1)

proportion = ...

(ii) Estimate the proportion of the body that is the legs in an 18-year old person.

(1)

proportion = ...

(b) Describe two main changes in the proportions of the body between birth and 18 years.

(2)

1 .....

.....

2 .....

.....

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9. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

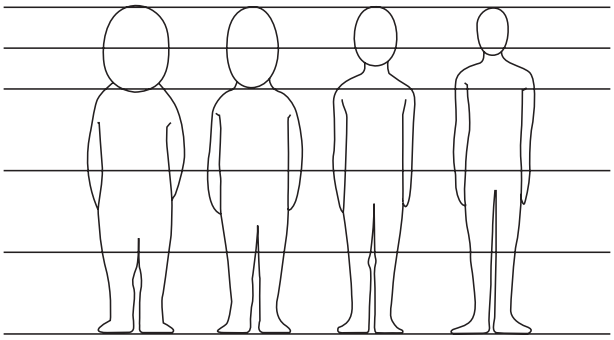
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10. Question 4 · part a.ii · 1 mark

June 2019 Paper 2R

Further practice · Attempt part a.ii only.

4 The diagram shows the changes in the proportions of different parts of the body of a person over a period of 18 years.



newborn 2 years 7 years 18 years

(a) (i) Estimate the proportion of the body that is the head in a newborn baby.

(1)

proportion = ...

(ii) Estimate the proportion of the body that is the legs in an 18-year old person.

(1)

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(2)

1 .....

.....

2 .....

.....

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10. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

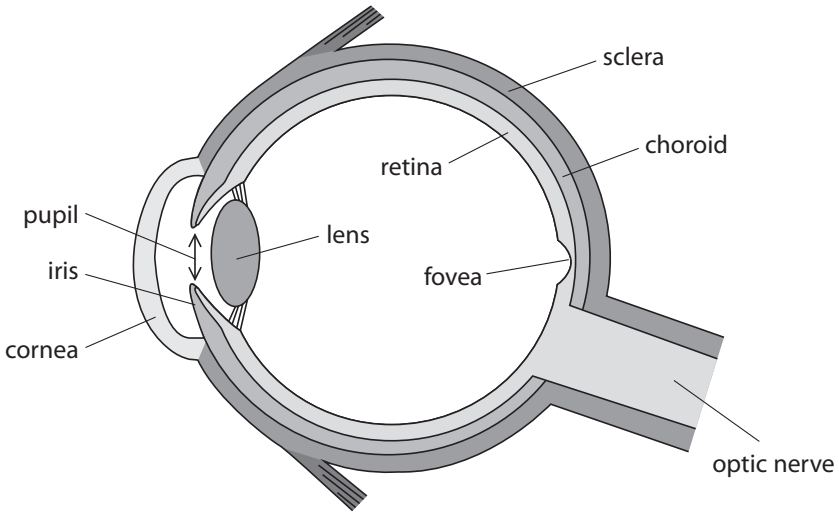
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11. Question 3 · part b · 1 mark

January 2020 Paper 2

Further practice · Attempt part b only.

3 (a) The diagram shows the human eye, with some structures labelled.



(i) Which structure helps to focus light at the back of the eye? **(1)**

- ☒ **A** fovea
- ☒ **B** iris
- ☒ **C** lens
- ☒ **D** pupil

(ii) The box lists structures of the eye. **(3)**

sclera	retina	choroid	iris	pupil

Use words from the box to complete the passage about the eye.

The ..... contains receptor cells that are sensitive to light.

In bright light the ..... gets smaller to control the amount of light entering the eye.

The ..... contains blood vessels that deliver oxygen to receptor cells.

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11. Question 3 · part b · 1 mark

January 2020 Paper 2

Further practice · Attempt part b only.

(b) The photomicrograph gives the thickness of the two outer layers of the eye.



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Estimate the thickness of the retina.

(1)

thickness of the retina = ... µm

(c) The optic nerve contains sensory neurones.

(i) Describe the function of the sensory neurones in the optic nerve.

(2)

.....

.....

.....

.....

(ii) Glaucoma is a disease that damages the optic nerve.

Explain why damage to the optic nerve will affect vision.

(2)

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(Total for Question 3 = 9 marks)

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11. Question 3 · part b · 1 mark · Working and answer

Working and answer

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